

SUBNETTING CHEATSHEET

Complete Reference for Network Engineers & Students

What's Inside:

- IP Address Classes
- CIDR Notation
- Subnet Mask Quick Table
- Usable Hosts Formula
- Wildcard Mask
- VLSM Concept
- Practice Tricks

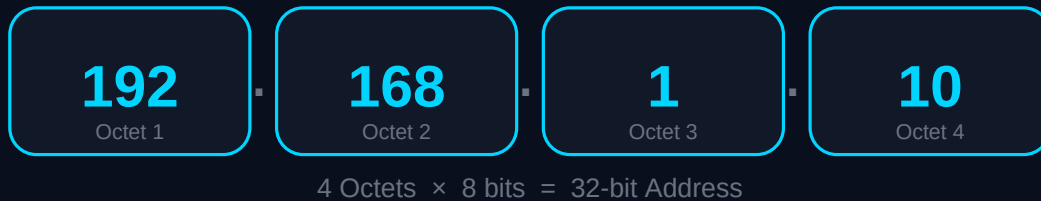
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By Akhilesh Chaudhari

@akhilesh.webdev | akhileshchaudhari.com

IP ADDRESS BASICS

IPv4 Structure & Binary



Binary: 11000000.10101000.00000001.00001010

Each octet = 8 bits | Range: 0–255

Class	Range	Default Mask	Usage
A	1–126	255.0.0.0 /8	Large Networks
B	128–191	255.255.0.0 /16	Medium Networks
C	192–223	255.255.255.0 /24	Small Networks
D	224–239	—	Multicast
E	240–255	—	Reserved/Research

■ **Trick:**

127.x.x.x = Loopback | 169.254.x.x = APIPA (No DHCP found)

SUBNET MASK & CIDR

The /24 Notation Explained

/24

255.255.255.0

256 IPs
254 Usable

/16

255.255.0.0

65536 IPs
65534 Usable

/8

255.0.0.0

16M IPs
16M Usable

How CIDR prefix works:

/24 means → 24 bits are NETWORK + 8 bits are HOST

11111111.11111111.11111111.00000000

← 24 ones (network) → ← 8 zeros (host) →

CIDR	Subnet Mask	Hosts	Subnets from /24
/30	255.255.255.252	2	64
/29	255.255.255.248	6	32
/28	255.255.255.240	14	16
/27	255.255.255.224	30	8
/26	255.255.255.192	62	4
/25	255.255.255.128	126	2
/24	255.255.255.0	254	1

SUBNET MASK QUICK TABLE

CIDR /1 to /32 Reference

CIDR	Subnet Mask	Usable Hosts
/8	255.0.0.0	16,777,214
/9	255.128.0.0	8,388,606
/10	255.192.0.0	4,194,302
/11	255.224.0.0	2,097,150
/12	255.240.0.0	1,048,574
/13	255.248.0.0	524,286
/14	255.252.0.0	262,142
/15	255.254.0.0	131,070
/16	255.255.0.0	65,534
/17	255.255.128.0	32,766
/18	255.255.192.0	16,382
/19	255.255.224.0	8,190
/20	255.255.240.0	4,094
/21	255.255.248.0	2,046
/22	255.255.252.0	1,022
/23	255.255.254.0	510
/24	255.255.255.0	254
/25	255.255.255.128	126
/26	255.255.255.192	62
/27	255.255.255.224	30
/28	255.255.255.240	14
/29	255.255.255.248	6
/30	255.255.255.252	2
/31	255.255.255.254	0 (P2P)
/32	255.255.255.255	1 (Host)

USABLE HOSTS FORMULA

Calculate Without Calculator!

Usable Hosts =

$$2^{(\text{Host Bits})} - 2$$

-2 because: 1 Network Address + 1 Broadcast Address are reserved

Worked Examples:

/24

32-24=8 host bits
 $2^8 - 2$

254 hosts

/26

32-26=6 host bits
 $2^6 - 2$

62 hosts

/28

32-28=4 host bits
 $2^4 - 2$

14 hosts

/30

32-30=2 host bits
 $2^2 - 2$

2 hosts

Number of Subnets: $2^{(\text{Subnet Bits Borrowed})}$

e.g. Borrow 3 bits from /24 → $2^3 = 8$ subnets, each /27 with 30 hosts

WILDCARD MASK

Used in ACLs & OSPF Config

Wildcard Mask =

255.255.255.255 - Subnet Mask

Subnet Mask	Wildcard Mask	CIDR
255.255.255.0	0.0.0.255	/24
255.255.255.128	0.0.0.127	/25
255.255.255.192	0.0.0.63	/26
255.255.255.224	0.0.0.31	/27
255.255.255.240	0.0.0.15	/28
255.255.255.248	0.0.0.7	/29
255.255.255.252	0.0.0.3	/30
255.255.0.0	0.0.255.255	/16
255.0.0.0	0.255.255.255	/8

Cisco ACL Example:

```
access-list 10 permit 192.168.1.0 0.0.0.255
```

→ Allows all 254 hosts in 192.168.1.0/24 subnet

→ '0' in wildcard = must match | '255' = any value allowed

VLSM & PRIVATE IPs

Variable Length Subnet Masking

What is VLSM?

VLSM allows different subnets to have different sizes.
Instead of wasting IPs, you assign exactly what's needed.
e.g. WAN link needs only 2 hosts → use /30 (not /24!)

VLSM Example: 192.168.1.0 /24 divided into:

Dept / Use	Hosts Needed	Assigned Subnet	Mask
Sales	50 hosts	192.168.1.0/26	/26 → 62 usable
HR	25 hosts	192.168.1.64/27	/27 → 30 usable
IT	12 hosts	192.168.1.96/28	/28 → 14 usable
WAN Link	2 hosts	192.168.1.112/30	/30 → 2 usable

Private IP Ranges (RFC 1918):

Class	Range	CIDR	Common Use
A	10.0.0.0 – 10.255.255.255	/8	Large Enterprises
B	172.16.0.0 – 172.31.255.255	/12	Medium Networks
C	192.168.0.0 – 192.168.255.255	/16	Home / SOHO

Special Addresses:

0.0.0.0	→ Default Route / Any address
127.0.0.1	→ Loopback (localhost)
169.254.x.x	→ APIPA — DHCP failure fallback
255.255.255.255	→ Limited Broadcast

SUBNETTING TRICKS

Shortcuts to Solve Faster

1

Magic Number Trick

256 - Subnet Mask Octet = Block Size

e.g. /26 → 256-192=64. Subnets: 0,64,128,192

2

Quick /24 Subdivision

Hosts per subnet = $2^{\text{host_bits}} - 2$

Subnets needed → find next power of 2

3

CIDR to Hosts (Mental Math)

/30=2, /29=6, /28=14, /27=30, /26=62, /25=126

Each step doubles: 2→6→14→30→62→126→254

4

Network & Broadcast Fast Calc

Network = First IP of block | Broadcast = Next block - 1

e.g. /26: block 64 → Network=x.x.x.64, Broadcast=x.x.x.127

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